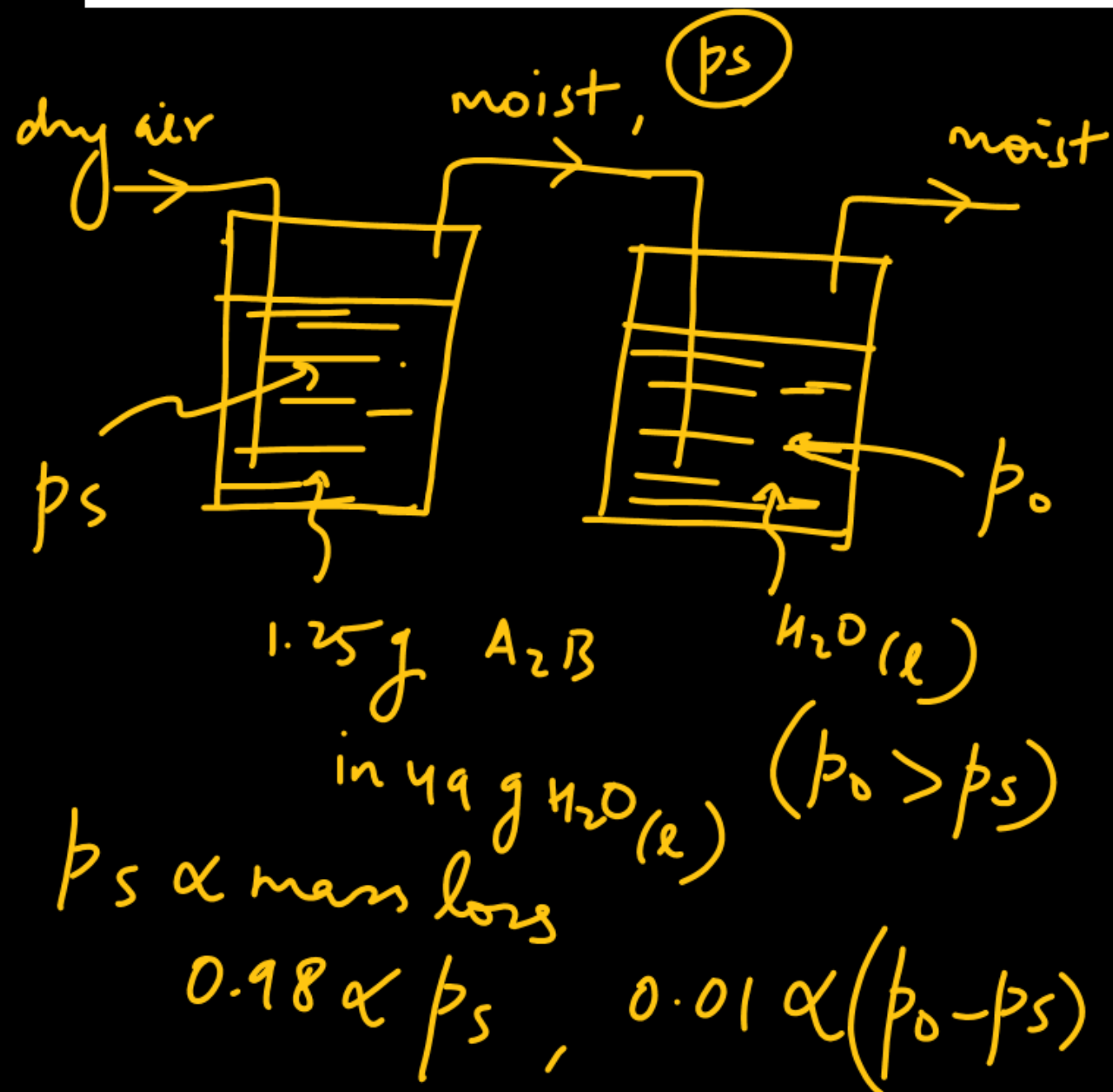


A current of dry air was passed through a series of bulbs containing 1.25 g of a solute A_2B (molar mass = 90 g/mol) in 49 g of water and then through pure water. The loss in weight of the former series of bulbs was 0.98 g and in the later series 0.01 g. The percentage dissociation of solute is

Ans: 50%



$$\frac{p_0 - p_s}{p_s} = \frac{i n}{N}$$

$$\frac{0.01}{0.98} = \frac{(1 + 2\alpha) \times \left(\frac{1.25}{90}\right)}{\left(\frac{49}{18}\right)}$$

$$\frac{1}{98} = \frac{(1 + 2\alpha) \times 1.25}{5 \times 49}$$

$$1 + 2\alpha = \frac{5}{2.5} = 2$$

$$\alpha = 0.5, \text{ \% dissociation of } A_2B = 50\% \text{ Ans.}$$

$$A_2B \rightleftharpoons 2A^+ + B^{2-}$$

$$1 - \alpha \quad 2\alpha \quad \alpha$$

$$\bar{i} = (1 + 2\alpha)$$

Ostwald and Walker Method

The value of 'x' g of a non-electrolytic compound ($M=200$) is dissolved in 1 L of 0.05 M NaCl aqueous solution. The osmotic pressure of this solution is found to be 4.92 atm at 27°C . Assume complete dissociation of NaCl and ideal behaviour for solution. The value of 'x' is ($R = 0.082 \text{ L-atm mol}^{-1}\text{K}^{-1}$)

Ans: 20 g

Solⁿ: $\pi = 4.92 \text{ atm}$, $T = 300 \text{ K}$, $R = 0.082 \text{ lit atm mol}^{-1}\text{K}^{-1}$.

$$\pi = iCRT$$

$$= (i_1c_1 + i_2c_2) \times RT$$

$$4.92 = \left\{ 1 \times \frac{(x/200)}{1} + (2 \times 0.05) \right\} \times 0.082 \times 300$$

$$4.92 = \left(\frac{x}{200} + 0.1 \right) \times 24.6$$

$$\frac{x}{200} + 0.1 = 0.2 \Rightarrow \frac{x}{200} = 0.1 \Rightarrow \underline{x = 20 \text{ g}} \quad \text{Q}$$

$$\underline{i_{\text{NaCl}} = 2}$$

→ on adding 'x' in NaCl aq. solⁿ, volume is taken constant i.e. 1 lit.

Equal masses of two aqueous solutions, one of 3.6g glucose per 100g of water and the other of 3.6g urea per 100g of water are mixed. If the freezing point of resulting solution is $-x^{\circ}\text{C}$, then the value of '1000x' is (K_f of water = $1.86 \text{ K Kg mol}^{-1}$)

Solⁿ: $3.6 \text{ g } \underbrace{\text{C}_6\text{H}_{12}\text{O}_6}_{180 \text{ g/mol}} + 3.6 \text{ g } \text{NH}_2\text{CONH}_2$, Total mass of solⁿ = $100 + 100 = 200 \text{ g}$.

$$n_{\text{C}_6\text{H}_{12}\text{O}_6} = \frac{3.6}{180} = 2 \times 10^{-2} \text{ mol}; i = 1$$

$$n_{\text{urea}} = \frac{3.6}{60} = 6 \times 10^{-2} \text{ mol}; i = 1$$

$$\Delta T_f = \left(T_f \right)_{\text{water}} - \left(T_f \right)_{\text{solution}} = i \times K_f \times m$$

$$0 - (-x) = 1 \times 1.86 \times \frac{(2 \times 10^{-2} + 6 \times 10^{-2}) \text{ mol}}{\left(\frac{192.7}{1000} \text{ Kg} \right)}$$

$$\Rightarrow x = 1.86 \times \frac{80}{192.7} = 0.772^{\circ}\text{C}$$

'1000x' = 772 An

$W_{\text{solvent}} = 200 \text{ g} - 3.6 - 3.6 = 200 - 7.2 = 192.7 \text{ g}$

सादधान

An amount of 1 mole of a non-volatile solid is dissolved in 200 moles of water. The solution is cooled to a temperature 'T' K (lower than the freezing point of the solution) to cause ice formation. After the removal of ice, the remaining solution is heated to 373 K, where the vapour pressure of solution is observed to be 740 mm Hg. Identify the correct information(s) from the following (K_f of water = 2.0 K·kg/mol).

(A) 163 moles of ice is formed at T K. ✓

(B) $T_K = \left(273 - \frac{2000}{37 \times 18} \right) K$ ✓

(C) Freezing point of the original solution is $-\frac{10}{18}^\circ C$. ✓

(D) Relative lowering of vapour pressure of the final solution is $\frac{1}{38}$. ✓

Ans: A, B, C, D

$$RLVP = \frac{p_0 - p_s}{p_0}$$

$$= \frac{760 - 740}{760}$$

$$= \frac{1}{38} \text{ A.}$$

Solⁿ. 1 mol solute, 200 mol water
 $p_s = 740$, $(p_0) = 760$ torr.
 $\rightarrow x$ mol frozen as $H_2O(s)$

$$\frac{p_0 - p_s}{p_s} = \frac{i n}{N} \Rightarrow \frac{760 - 740}{740} = \frac{1 \times 1}{(200 - x)}$$

$$\frac{2}{74} = \frac{1}{200 - x} \Rightarrow 200 - x = 37$$

$x = 163$ mol frozen as ice.

$$\Delta T_f = i \times K_f \times m$$

$$273 - T = 1 \times 2 \times \frac{1 \text{ mol}}{\left(\frac{37 \times 18}{1000} \right)}$$

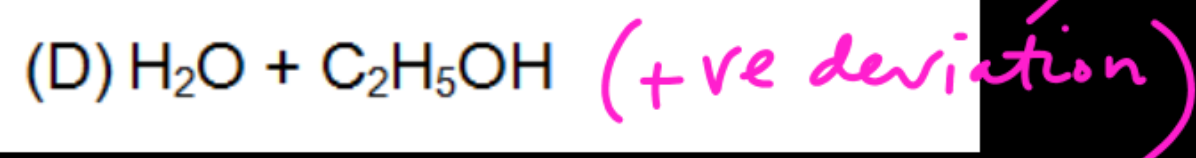
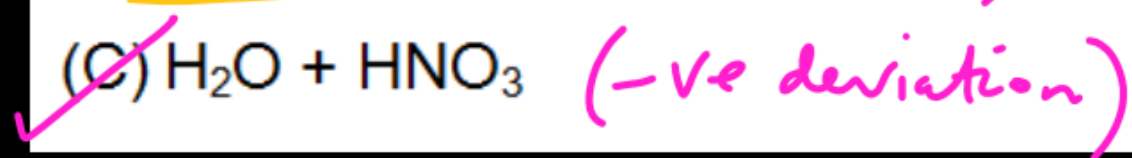
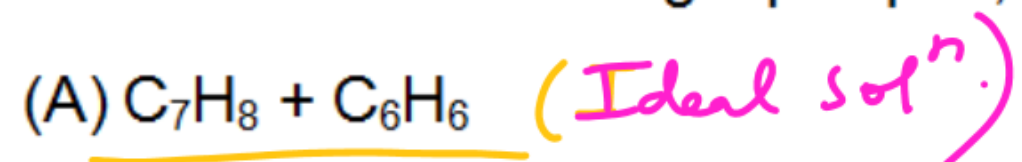
$$T = \left(273 - \frac{1000 \times 2}{37 \times 18} \right) K.$$

$$273 - (T_f)_{\text{original}} = 1 \times 2 \times \frac{1}{\left(\frac{200 \times 18}{1000} \right)}$$

$$(T_f)_{\text{original}} = \left(273 - \frac{10}{18} \right) K$$

$$(t_f)_{\text{original}} = -\frac{10}{18}^\circ C \text{ Q}$$

In which of the following liquid pair, there is contraction in volume on mixing?



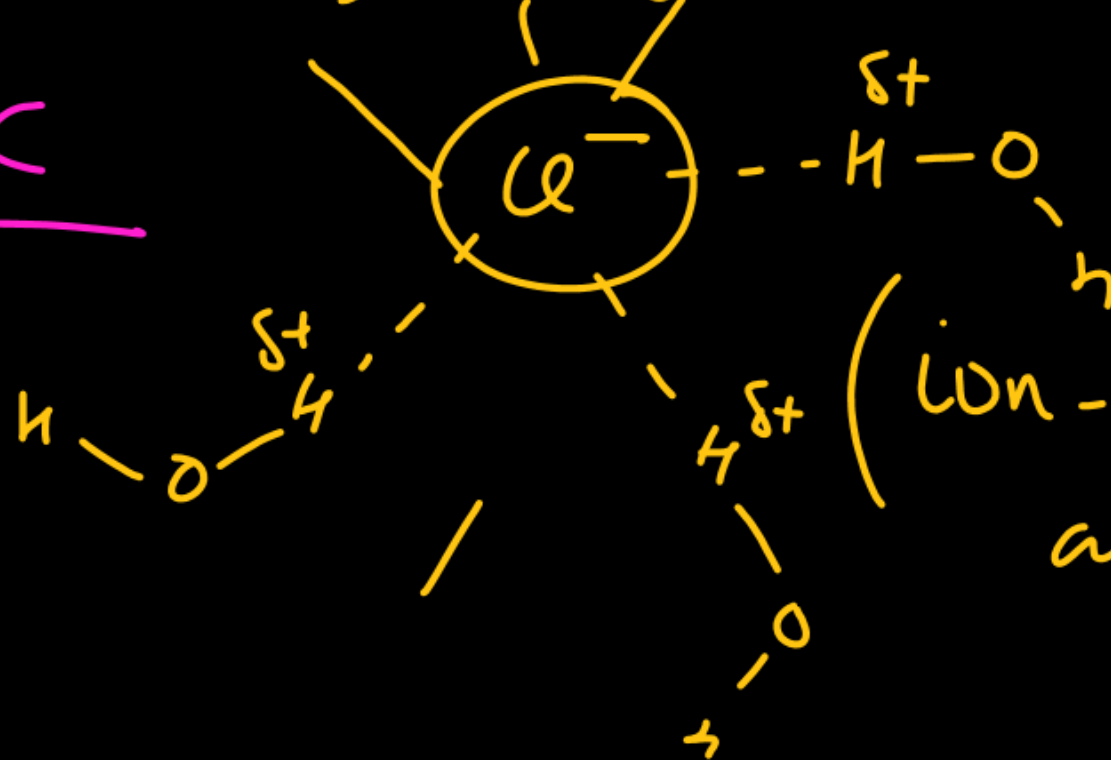
Solⁿ: For non-ideal solution,

+ve deviation from Raoult's law.

$p_s > p_{\text{Raoult's}}$
Volume expands.

-ve deviation ($p_s < p_{\text{Raoult's}}$)
Volume contraction.

Ans: B, C



(ion-dipole attractions are stronger than dipole-dipole attⁿ).

Imp

The vapour pressure of a saturated solution of sparingly soluble salt (XCl_3) was 17.20 mm Hg at 27°C . If the vapour pressure of pure water is 17.25 mm Hg at 27°C , then what is the solubility of the sparingly soluble salt XCl_3 in mole per litre?

(A) 4.04×10^{-2}

(B) 8.08×10^{-2}

(C) 2.02×10^{-2}

(D) 4.04×10^{-3}

Solⁿ: $\text{XCl}_3(\text{s}) \rightleftharpoons \text{X}^{3+} + 3\text{Cl}^-$, $i = 4$ (sparingly soluble salts are considered to be strong electrolytes)

$(S) \text{ mol/lit}$ S $3S$

$p_s = 17.2 \text{ mm}$, $p_0 = 17.25 \text{ mm of Hg}$

$$\frac{p_0 - p_s}{p_s} = \frac{i n}{N}$$
$$\frac{17.25 - 17.2}{17.2} = \frac{4 \times S}{\left(\frac{1000}{18} \text{ mol}\right)}$$
$$\frac{0.050}{17.2} = \frac{4 \times 18 \times S}{1000}$$
$$S = \frac{50 \times 12.5}{17.2 \times 4 \times 18} = \frac{12.5}{17.2 \times 18}$$
$$= 4.04 \times 10^{-2} \text{ mol/lit}$$

A dilute solution contains n mole of solute 'A' in 1 kg of a solvent with molal elevation constant K_b . The solute A undergoes dimerization as $2A \rightleftharpoons A_2$. If ΔT_b is the elevation in boiling point of a given solution and molality = molarity, the equilibrium constant K_c for dimer formation is

(A) $\frac{K_b(nK_b - \Delta T_b)}{(2\Delta T_b - nK_b)}$

(B) $\frac{K_b(nK_b - \Delta T_b)}{(2\Delta T_b - nK_b)^2}$

(C) $\frac{K_b n \Delta T_b}{(2\Delta T_b - nK_b)^2}$

(D) $\frac{(nK_b - \Delta T_b)}{(n\Delta T_b - 2K_b)^2}$

$K_c = \frac{(nK_b - \Delta T_b)}{\left(n - 2n + \frac{2\Delta T_b}{K_b}\right)^2}$

Molality = molarity
 $\frac{x}{V_{\text{solution in lit}}} = \frac{x \text{ mol}}{1 \text{ kg}}$

Solⁿ: $A \rightarrow n \text{ mol, 1 kg solvent, } K_b = \frac{(nK_b - \Delta T_b)}{K_b \left(\frac{2\Delta T_b}{K_b} - n\right)^2}$ $V = 1 \text{ lit}$

$2A \rightleftharpoons A_2, K_c = \frac{[A_2]_{eq}}{[A]_{eq}^2} = \frac{\left(\frac{x}{2}\right)/1}{\left(\frac{n-x}{1}\right)^2} = \frac{x}{2(n-x)^2}$ — (1)

$t=0, n$

$t=t_{eq}, n-x \quad x/2$

$i = \frac{(n_T)_f}{(n_T)_i} = \frac{(n-x + x/2)}{n} = \frac{n - x/2}{n} = \frac{(2n-x)}{2n}$

$\Delta T_b = i \times K_b \times m$

$\Delta T_b = \left(1 - \frac{x}{2n}\right) K_b \times \frac{n}{1}$

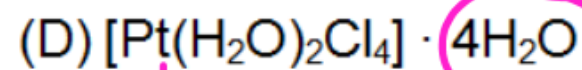
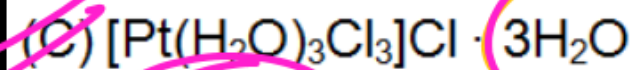
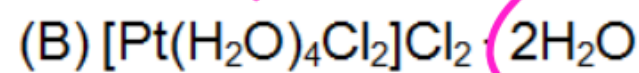
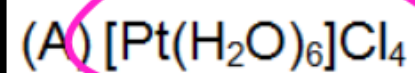
$1 - \frac{x}{2n} = \frac{\Delta T_b}{K_b \times n}$

$\Rightarrow \frac{x}{2n} = 1 - \frac{\Delta T_b}{K_b \times n}$

$x = 2n - \frac{2\Delta T_b}{K_b}$
 $x = \frac{2nK_b - 2\Delta T_b}{K_b}$

$K_c = \frac{K_b(nK_b - \Delta T_b)}{(2\Delta T_b - nK_b)^2}$

$\text{PtCl}_4 \cdot 6\text{H}_2\text{O}$ can exist as a hydrated complex. 1.0 molal aqueous solution has depression in freezing point of 3.72°C . Assume 100% ionization and K_f of water = $1.86^\circ\text{C mol}^{-1}\text{kg}$. The complex is



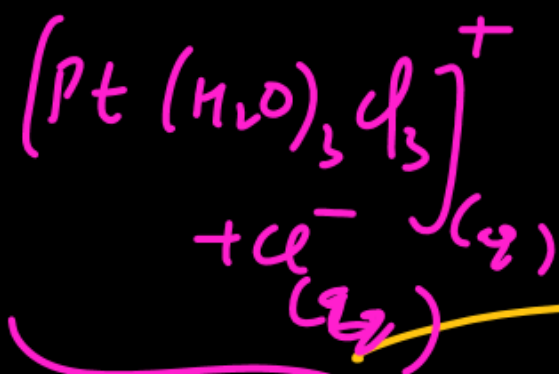
Solⁿ:

1 molal solution, $\Delta T_f = 3.72^\circ\text{C}$; $K_f = 1.86^\circ\text{C mol}^{-1}\text{kg}$.

$$\Delta T_f = i \times K_f \times m$$

$$3.72 = i \times 1.86 \times 1$$

$$i = 2$$



2 particles.

Two liquids 'A' and 'B' form a solution which shows significant positive deviation from Raoult's law. The normal boiling point of pure liquid 'A' is higher than that of 'B'. Which of the following information(s) related with the solution must be incorrect?

- (A) Normal boiling point of the solution of any composition will always be less than that of 'A'. ✗
- (B) Azeotropic mixture of 'A' and 'B' will have normal boiling point lower than that of 'B'. ✗
- ✓ (C) If a small amount of liquid 'B' is added in the liquid solution, the normal boiling point of solution will always decrease. ✓
- ✓ (D) On distillation of the solution at constant pressure of 1 atm, the distillate will have pure liquid 'Y' and the residual liquid will be pure liquid 'X'. ✗

→ (B.P) azeotrope with +ve deviation is less than B.P. of individual components.

